

YIXI CAI

Curriculum Vitae

Department of Robotics, Perception and Learning

KTH Royal Institute of Technology

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Work Experience

KTH Royal Institute of Technology

Postdoctoral Fellow, Digital Future Fellowship

Supervisor: Prof. Patric Jensfelt and Dr. Olov Andersson

Stockholm, Sweden

Jan, 2025 – Present

Education

University of Hong Kong

Ph.D., Robotics

Supervisor: Prof. Fu Zhang and Prof. James Lam

Hong Kong SAR, China

Sept, 2020 – Sept, 2024

Beihang University

B.Eng, Automation

Beijing, China

Sept, 2016 – July, 2020

Featured Publication

*: Equal Contribution, ‡: Corresponding Author, †: Project Leader

[RAL'2026] Waqas Ali, **Yixi Cai**[‡], Patric Jensfelt, and Thien-Minh Nguyen. Probper-lilo: Probabilistic persistency modeling for life-long mapping. *IEEE Robotics and Automation Letters*, volume 11, pages 2530–2537. IEEE, 2026. [Paper].

[RSS'2025] Yuhan Xie, **Yixi Cai**[†], Yinqiang Zhang, Lei Yang, and Jia Pan. GauSS-MI: Gaussian splatting shannon mutual information for active 3d reconstruction. In *Robotics: Science and Systems*, 2025. [Paper].

[TRO'2023] **Yixi Cai**^{*}, Fanze Kong^{*}, Yunfan Ren, Fangcheng Zhu, Jiarong Lin, and Fu Zhang. Occupancy grid mapping without ray-casting for high-resolution lidar sensors. *IEEE Transactions on Robotics*, volume 40, pages 172–192, 2024. [Paper].

[TRO'2022] Wei Xu^{*}, **Yixi Cai**^{*}, Dongjiao He, Jiarong Lin, and Fu Zhang. FAST-LIO2: Fast direct lidar-inertial odometry. *IEEE Transactions on Robotics*, volume 38, pages 2053–2073. IEEE, 2022. [Paper] (**ESI Highly Cited Paper, Rank 2nd in Most Cited Papers in LiDAR-Inertial Odometry, Top 10 TRO Popular Paper**).

[SR'2025] Yunfan Ren, Fangcheng Zhu, Guozheng Lu, **Yixi Cai**, Longji Yin, Fanze Kong, Jiarong Lin, Nan Chen, and Fu Zhang. Safety-assured high-speed navigation for mavs. *Science Robotics*, volume 10, page eado6187, 2025. [Paper].

[SR'2023] Nan Chen, Fanze Kong, Wei Xu, **Yixi Cai**, Haotian Li, Dongjiao He, Youming Qin, and Fu Zhang. A self-rotating, single-actuated uav with extended sensor field of view for autonomous navigation. *Science Robotics*, volume 8, page eade4538, 2023. [Paper].

Fellowships & Awards

Digital Future Postdoctoral Fellowship, 2025-2026

RSS Pioneer in Robotics: Science and Systems 2024 (only 1 successful applicant from China)

Best Paper Award Finalist in IEEE/ASME Transactions on Mechatronics, 2023

Y S and Christabel Lung Postgraduate Scholarship for Engineering Students 2020-2021.
University Postgraduate Fellowships for the academic year 2020-21.

Academic Service

Associate Editor, Robotics and Automation Letters (RA-L)
RSS Pioneer Program Committee 2025

Invited Talks

Towards Efficient LiDAR Mapping for Robotics <i>Smart Robotics Lab, Technical University of Munich</i>	Online 2025.02.23
Towards Efficient LiDAR Mapping for Robotics <i>Southern University of Science and Technology</i>	Onsite 2024.09.20
Towards Spatial Perception for Robotics <i>Robot Perception and Learning Lab, University College London</i>	Onsite 2025.06.03
Towards Spatial Perception for Robotics <i>Oxford Robotics Institute, University of Oxford</i>	Onsite 2025.06.06
Towards Efficient Mapping for Robotics <i>Global College, Shanghai Jiao Tong University</i>	Onsite 2025.10.25

Teaching

MECH3433 Robotics, drones and autonomous ground vehicles <i>Teaching Assistant</i>	Spring Semester 2020-2024
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Co-supervisory

Bingyang Zhou <i>University of Hong Kong</i>	Master of Philosophy 2023-2025
Ziliang Miao <i>University of Hong Kong</i>	Master of Philosophy 2023-2025
Ajinkya Khoche <i>KTH Royal Institute of Technology</i>	Doctor of Philosophy 2025-Present
Waqas Ali <i>KTH Royal Institute of Technology</i>	Doctor of Philosophy 2025-Present
Alejandro Sánchez Roncero <i>KTH Royal Institute of Technology</i>	Doctor of Philosophy 2025-Present

Full Publication Lists

- [RAL'2026] Waqas Ali, **Yixi Cai**[‡], Patric Jensfelt, and Thien-Minh Nguyen. Probper-lilo: Probabilistic persistency modeling for life-long mapping. *IEEE Robotics and Automation Letters*, volume 11, pages 2530–2537. IEEE, 2026. [Paper].
- [NCE'2025] Yuying Zou, Haotian Li, Yunfan Ren, Fanze Kong, Wei Xu, Yihang Li, **Yixi Cai**, and Fu Zhang. Enabling multirotor uavs to perch, land and detach with standard propeller guards. *Communications Engineering*, volume 4, page 185. Nature Publishing Group UK London, 2025. [Paper].
- [RAL'2025] Bingyang Zhou, Chunran Zheng, Ziming Wang, Fangcheng Zhu, **Yixi Cai**, and Fu Zhang. Fast-livo2 on resource-constrained platforms: Lidar-inertial-visual odometry with efficient memory and computation. *IEEE Robotics and Automation Letters*, volume 10, pages 7931–7938, 2025. [Paper].
- [NIPS'2025] Qingwen Zhang, Xiaomeng Zhu, Yushan Zhang, **Yixi Cai**, Olov Andersson, and Patric Jensfelt. Deltaflow: An efficient multi-frame scene flow estimation method. *Advances in Neural Information Processing Systems*, 2025. **Spotlight** [Paper].
- [RSS'2025] Yuhan Xie, **Yixi Cai**[†], Yinqiang Zhang, Lei Yang, and Jia Pan. GauSS-MI: Gaussian splatting shannon mutual information for active 3d reconstruction. In *Robotics: Science and Systems*, 2025. [Paper].
- [RAL'2025] Hairuo Wei, Rundong Li, **Yixi Cai**, Chongjian Yuan, Yunfan Ren, Zuhao Zou, Huajie Wu, Chunran Zheng, Shunbo Zhou, Kaiwen Xue, and Fu Zhang. Large-scale multi-session point-cloud map merging. *IEEE Robotics and Automation Letters*, volume 10, pages 88–95, 2025. [Paper].
- [SR'2025] Yunfan Ren, Fangcheng Zhu, Guozheng Lu, **Yixi Cai**, Longji Yin, Fanze Kong, Jiarong Lin, Nan Chen, and Fu Zhang. Safety-assured high-speed navigation for mavs. *Science Robotics*, volume 10, page eado6187, 2025. [Paper].
- [TMECH'2025] Yunfan Ren, **Yixi Cai**, Haotian Li, Nan Chen, Fangcheng Zhu, Longji Yin, Fanze Kong, Rundong Li, and Fu Zhang. A survey on lidar-based autonomous aerial vehicles. *IEEE/ASME Transactions on Mechatronics*, pages 1–17, 2025. [Paper].
- [ICCV'2025] Ziliang Miao, Runjian Chen, **Yixi Cai**, Buwei He, Wenquan Zhao, Wenqi Shao, Bo Zhang, and Fu Zhang. Temporal overlapping prediction: A self-supervised pre-training method for lidar moving object segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 26653–26663, October 2025. [Paper].
- [NC'2025] Wenyi Liu, Yunfan Ren, Rui Guo, Vickie WW Kong, Anthony SP Hung, Fangcheng Zhu, **Yixi Cai**, Huajie Wu, Yuying Zou, and Fu Zhang. Slope inspection under dense vegetation using lidar-based quadrotors. *Nature Communications*, volume 16, page 7411. Nature Publishing Group UK London, 2025. [Paper].
- [ICRA'2025] Jianheng Liu, Chunran Zheng, Yunfei Wan, Bowen Wang, Yixi Cai, and Fu Zhang. Neural surface reconstruction and rendering for lidar-visual systems. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2034–2040. IEEE, 2025. [Paper].
- [IROS'2025] Yisheng Li, Longji Yin, Yixi Cai, Jianheng Liu, Fangcheng Zhu, Mingpu Ma, Siqi Liang, Haotian Li, and Fu Zhang. Efficient swept volume-based trajectory generation for arbitrary-shaped ground robot navigation. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2076–2083. IEEE, 2025. [Paper].
- [TRO'2025] Song Li, Songnan Bai, Ruihan Jia, **Yixi Cai**, Runze Ding, Yu Shi, Fu Zhang, and Pakpong Chirattananon. A high-payload robotic hopper powered by bidirectional thrusters. *IEEE Transactions on Robotics*. IEEE, 2025. [Paper].

- [IJRR'2025] Rundong Li, Zheng Liu, Hairuo Wei, **Yixi Cai**, Haotian Li, and Fu Zhang. Efficient and distributed large-scale point cloud bundle adjustment via majorization-minimization. *The International Journal of Robotics Research*. SAGE Publications Sage UK: London, England, 2025. [Paper].
- [ICRA'2025] Rundong Li, Xiyuan Liu, Haotian Li, Zheng Liu, Jiarong Lin, Yixi Cai, and Fu Zhang. Lvba: Lidar-visual bundle adjustment for rgb point cloud mapping. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2041–2047. IEEE, 2025. [Paper].
- [TRO'2024] Fangcheng Zhu, Yunfan Ren, Longji Yin, Fanze Kong, Qingbo Liu, Ruize Xue, Wenyi Liu, **Yixi Cai**, Guozheng Lu, Haotian Li, et al. Swarm-lid2: Decentralized, efficient lidar-inertial odometry for uav swarms. *IEEE Transactions on Robotics*. IEEE, 2024. [Paper].
- [TRO'2023] **Yixi Cai***, Fanze Kong*, Yunfan Ren, Fangcheng Zhu, Jiarong Lin, and Fu Zhang. Occupancy grid mapping without ray-casting for high-resolution lidar sensors. *IEEE Transactions on Robotics*, volume 40, pages 172–192, 2024. [Paper].
- [IROS'2024] Yunfan Ren, **Yixi Cai**, Fangcheng Zhu, Siqi Liang, and Fu Zhang. Rog-map: An efficient robocentric occupancy grid map for large-scene and high-resolution lidar-based motion planning. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 8119–8125. IEEE, 2024. [Paper].
- [TRO'2024] Guozheng Lu, Yunfan Ren, Fangcheng Zhu, Haotian Li, Ruize Xue, **Yixi Cai**, Ximin Lyu, and Fu Zhang. Autonomous tail-sitter flights in unknown environments. *IEEE Transactions on Robotics*. IEEE, 2024. [Paper].
- [IJRR'2024] Haotian Li, Yuying Zou, Nan Chen, Jiarong Lin, Xiyuan Liu, Wei Xu, Chunran Zheng, Rundong Li, Dongjiao He, Fanze Kong, **Yixi Cai**, et al. Mars-lvig dataset: A multi-sensor aerial robots slam dataset for lidar-visual-inertial-gnss fusion. *The International Journal of Robotics Research*. SAGE Publications Sage UK: London, England, 2024. [Paper].
- [CASE'2023] Yuying Zou, Haotian Li, Yunfan Ren, Wei Xu, Yihang Li, **Yixi Cai**, Shenji Zhou, and Fu Zhang. Perch a quadrotor on planes by the ceiling effect. In *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)*, pages 1–7. IEEE, 2023. [Paper].
- [IJRR'2023] Guozheng Lu, **Yixi Cai**, Nan Chen, Fanze Kong, Yunfan Ren, and Fu Zhang. Trajectory generation and tracking control for aggressive tail-sitter flights. *The International Journal of Robotics Research*, volume 43. SAGE Publications Sage UK: London, England, 2023. [Paper].
- [TRO'2023] Jiarong Lin, Chongjian Yuan, **Yixi Cai**, Haotian Li, Yunfan Ren, Yuying Zou, Xiaoping Hong, and Fu Zhang. Immesh: An immediate lidar localization and meshing framework. *IEEE Transactions on Robotics*, volume 39, pages 4312–4331, 2023. [Paper].
- [RAL'2023] Fanze Kong, Xiyuan Liu, Benxu Tang, Jiarong Lin, Yunfan Ren, **Yixi Cai**, Fangcheng Zhu, Nan Chen, and Fu Zhang. Marsim: A light-weight point-realistic simulator for lidar-based uavs. *IEEE Robotics and Automation Letters*, volume 8, pages 2954–2961. IEEE, 2023. [Paper].
- [SR'2023] Nan Chen, Fanze Kong, Wei Xu, **Yixi Cai**, Haotian Li, Dongjiao He, Youming Qin, and Fu Zhang. A self-rotating, single-actuated uav with extended sensor field of view for autonomous navigation. *Science Robotics*, volume 8, page eade4538. American Association for the Advancement of Science, 2023. [Paper].
- [TRO'2022] Wei Xu*, **Yixi Cai***, Dongjiao He, Jiarong Lin, and Fu Zhang. FAST-LIO2: Fast direct lidar-inertial odometry. *IEEE Transactions on Robotics*, volume 38, pages 2053–2073. IEEE, 2022. [Paper].
- [TCST'2022] Wei Xu, Dongjiao He, **Yixi Cai**, and Fu Zhang. Robots' state estimation and observability analysis based on statistical motion models. *IEEE Transactions on Control Systems Technology*, volume 30, pages 2030–2045. IEEE, 2022. [Paper].
- [TMECH'2022] Youming Qin, Nan Chen, **Yixi Cai**, Wei Xu, and Fu Zhang. Gemini ii: Design, modeling, and control of a compact yet efficient servoless bi-copter. *IEEE/ASME Transactions on Mechatronics*, volume 27, pages 4304–4315. IEEE, 2022. **Best Paper Award Finalist** [Paper].

[RAL'2021] Fanze Kong, Wei Xu, **Yixi Cai**, and Fu Zhang. Avoiding dynamic small obstacles with onboard sensing and computation on aerial robots. *IEEE Robotics and Automation Letters*, volume 6, pages 7869–7876. IEEE, 2021. [Paper].